Name: Nilav Ghosh Assignment: HW3

Assignment Statement:

After evaluating the data we deduced that the dataset is large, it has missing values, it is biased and has a large number of features, large number of observations and have unbalanced class. So we decided to work with models which scale well, handle non linear relationships, and can account for class imbalance. So the model order we followed:

Logistic regression (Different Permutations)- Nilav

XGBoost (Regular, Improved, Optimized, Deature Selected, SHAP)- Nilav

Naive Bayes - Nilav

Catbooster-Nilav

LightGBM-Nilav

Ensemble (XGBoost+CatBoost+LightGBM)-Nilav

Ada_boost - Ayoola

Neural Networks - Ayoola

Random Trees - Ayoola

Gradient boosting - Ayoola

Extra trees

This notebook contains Nilav's experiments.

In [7]:
```
#Importing necessary libraries
import pandas as pd
import numpy as np

from sklearn.model_selection import cross_val_score
from sklearn.metrics import accuracy_score
from sklearn.metrics import f1_score
from sklearn.metrics import roc_curve, auc

from sklearn.model_selection import GridSearchCV

from sklearn.model_selection import train_test_split

import seaborn as sns
import matplotlib.pyplot as plt
from matplotlib.colors import ListedColormap
%matplotlib inline
```

```
from numpy import loadtxt
import xgboost as xgb

from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC

# Going to use these 5 base models for the stacking
from sklearn.ensemble import (RandomForestClassifier, AdaBoostClassifier,
                              GradientBoostingClassifier, ExtraTreesClassifi
from sklearn.model_selection import KFold

from sklearn.naive_bayes import GaussianNB

import lightgbm as lgb
from sklearn.metrics import mean_squared_error

import matplotlib as mpl
from matplotlib import colors
from scipy import linalg
from sklearn import datasets
from sklearn.decomposition import PCA
from sklearn.discriminant_analysis import LinearDiscriminantAnalysis

from sklearn.model_selection import train_test_split, learning_curve
from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler, OneHotEncoder

from sklearn.dummy import DummyClassifier
from sklearn.linear_model import LogisticRegression

from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    roc_auc_score,
    classification_report,
    confusion_matrix
)
```

In [9]:
```
!kaggle --version
```

```
Kaggle CLI 2.2.4
```

In [10]:
```
import pandas as pd

train = pd.read_csv('./train_data.csv')
test = pd.read_csv('./test_data.csv')

train.output.value_counts()
```

Out[10]:
```
output
1    3873
0     711
Name: count, dtype: int64
```

In [11]:
```
# Read and load the training and test data
train = pd.read_csv("train_data.csv")
test = pd.read_csv("test_data.csv")

try:
    sample_solution = pd.read_csv("sample_solution.csv")
except:
    sample_solution = None

print("Train shape:", train.shape)
print("Test shape :", test.shape)

display(train.head())
```

```
Train shape: (4584, 670)
Test shape : (1732, 669)
```

	x1	x2	x3	x4	x5	x6	x7	x8	x9	x10
0	0	0	0	1	-323.106605	2.694366	-1.987520	261.092784	0.013704	0.000100
1	0	0	0	1	-376.084691	0.969696	-6.933765	355.311648	0.030292	-0.000153
2	0	0	0	0	91.955425	2.621643	-2.581162	51.357206	0.036668	-0.000104
3	0	0	0	1	-391.814586	1.866914	-2.510799	382.900317	0.007947	-0.000028
4	0	0	0	0	-363.823732	2.951346	-3.726368	330.527539	0.010074	-0.000004

5 rows × 670 columns

In [12]:
```
# Training Data information
train.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 4584 entries, 0 to 4583
Columns: 670 entries, x1 to output
dtypes: float64(474), int64(194), str(2)
memory usage: 23.4 MB
```

In [13]:
```
# Identify numerical and categorical columns

print("\nColumns:")
print(train.columns.tolist())

# Missing values:
print("\nMissing values:")
display(train.isnull().sum().sort_values(ascending=False).head(20))

# Unique values
print("\nUnique values:")
display(train.nunique().sort_values())
```

Columns:
['x1', 'x2', 'x3', 'x4', 'x5', 'x6', 'x7', 'x8', 'x9', 'x10', 'x11', 'x12',
'x13', 'x14', 'x15', 'x16', 'x17', 'x18', 'x19', 'x20', 'x21', 'x22', 'x23',
'x24', 'x25', 'x26', 'x27', 'x28', 'x29', 'x30', 'x31', 'x32', 'x33', 'x34',
'x35', 'x36', 'x37', 'x38', 'x39', 'x40', 'x41', 'x42', 'x43', 'x44', 'x45',
'x46', 'x47', 'x48', 'x49', 'x50', 'x51', 'x52', 'x53', 'x54', 'x55', 'x56',
'x57', 'x58', 'x59', 'x60', 'x61', 'x62', 'x63', 'x64', 'x65', 'x66', 'x67',
'x68', 'x69', 'x70', 'x71', 'x72', 'x73', 'x74', 'x75', 'x76', 'x77', 'x78',
'x79', 'x80', 'x81', 'x82', 'x83', 'x84', 'x85', 'x86', 'x87', 'x88', 'x89',
'x90', 'x91', 'x92', 'x93', 'x94', 'x95', 'x96', 'x97', 'x98', 'x99', 'x10
0', 'x101', 'x102', 'x103', 'x104', 'x105', 'x106', 'x107', 'x108', 'x109',
'x110', 'x111', 'x112', 'x113', 'x114', 'x115', 'x116', 'x117', 'x118', 'x11
9', 'x120', 'x121', 'x122', 'x123', 'x124', 'x125', 'x126', 'x127', 'x128',
'x129', 'x130', 'x131', 'x132', 'x133', 'x134', 'x135', 'x136', 'x137', 'x13
8', 'x139', 'x140', 'x141', 'x142', 'x143', 'x144', 'x145', 'x146', 'x147',
'x148', 'x149', 'x150', 'x151', 'x152', 'x153', 'x154', 'x155', 'x156', 'x15
7', 'x158', 'x159', 'x160', 'x161', 'x162', 'x163', 'x164', 'x165', 'x166',
'x167', 'x168', 'x169', 'x170', 'x171', 'x172', 'x173', 'x174', 'x175', 'x17
6', 'x177', 'x178', 'x179', 'x180', 'x181', 'x182', 'x183', 'x184', 'x185',
'x186', 'x187', 'x188', 'x189', 'x190', 'x191', 'x192', 'x193', 'x194', 'x19
5', 'x196', 'x197', 'x198', 'x199', 'x200', 'x201', 'x202', 'x203', 'x204',
'x205', 'x206', 'x207', 'x208', 'x209', 'x210', 'x211', 'x212', 'x213', 'x21
4', 'x215', 'x216', 'x217', 'x218', 'x219', 'x220', 'x221', 'x222', 'y1', 'y
2', 'y3', 'y4', 'y5', 'y6', 'y7', 'y8', 'y9', 'y10', 'y11', 'y12', 'y13', 'y
14', 'y15', 'y16', 'y17', 'y18', 'y19', 'y20', 'y21', 'y22', 'y23', 'y24',
'y25', 'y26', 'y27', 'y28', 'y29', 'y30', 'y31', 'y32', 'y33', 'y34', 'y35',
'y36', 'y37', 'y38', 'y39', 'y40', 'y41', 'y42', 'y43', 'y44', 'y45', 'y46',
'y47', 'y48', 'y49', 'y50', 'y51', 'y52', 'y53', 'y54', 'y55', 'y56', 'y57',
'y58', 'y59', 'y60', 'y61', 'y62', 'y63', 'y64', 'y65', 'y66', 'y67', 'y68',
'y69', 'y70', 'y71', 'y72', 'y73', 'y74', 'y75', 'y76', 'y77', 'y78', 'y79',
'y80', 'y81', 'y82', 'y83', 'y84', 'y85', 'y86', 'y87', 'y88', 'y89', 'y90',
'y91', 'y92', 'y93', 'y94', 'y95', 'y96', 'y97', 'y98', 'y99', 'y100', 'y10
1', 'y102', 'y103', 'y104', 'y105', 'y106', 'y107', 'y108', 'y109', 'y110',
'y111', 'y112', 'y113', 'y114', 'y115', 'y116', 'y117', 'y118', 'y119', 'y12
0', 'y121', 'y122', 'y123', 'y124', 'y125', 'y126', 'y127', 'y128', 'y129',
'y130', 'y131', 'y132', 'y133', 'y134', 'y135', 'y136', 'y137', 'y138', 'y13
9', 'y140', 'y141', 'y142', 'y143', 'y144', 'y145', 'y146', 'y147', 'y148',
'y149', 'y150', 'y151', 'y152', 'y153', 'y154', 'y155', 'y156', 'y157', 'y15
8', 'y159', 'y160', 'y161', 'y162', 'y163', 'y164', 'y165', 'y166', 'y167',
'y168', 'y169', 'y170', 'y171', 'y172', 'y173', 'y174', 'y175', 'y176', 'y17
7', 'y178', 'y179', 'y180', 'y181', 'y182', 'y183', 'y184', 'y185', 'y186',
'y187', 'y188', 'y189', 'y190', 'y191', 'y192', 'y193', 'y194', 'y195', 'y19
6', 'y197', 'y198', 'y199', 'y200', 'y201', 'y202', 'y203', 'y204', 'y205',
'y206', 'y207', 'y208', 'y209', 'y210', 'y211', 'y212', 'y213', 'y214', 'y21
5', 'y216', 'y217', 'y218', 'y219', 'y220', 'y221', 'y222', 'z1', 'z2', 'z
3', 'z4', 'z5', 'z6', 'z7', 'z8', 'z9', 'z10', 'z11', 'z12', 'z13', 'z14',
'z15', 'z16', 'z17', 'z18', 'z19', 'z20', 'z21', 'z22', 'z23', 'z24', 'z25',
'z26', 'z27', 'z28', 'z29', 'z30', 'z31', 'z32', 'z33', 'z34', 'z35', 'z36',
'z37', 'z38', 'z39', 'z40', 'z41', 'z42', 'z43', 'z44', 'z45', 'z46', 'z47',
'z48', 'z49', 'z50', 'z51', 'z52', 'z53', 'z54', 'z55', 'z56', 'z57', 'z58',
'z59', 'z60', 'z61', 'z62', 'z63', 'z64', 'z65', 'z66', 'z67', 'z68', 'z69',
'z70', 'z71', 'z72', 'z73', 'z74', 'z75', 'z76', 'z77', 'z78', 'z79', 'z80',
'z81', 'z82', 'z83', 'z84', 'z85', 'z86', 'z87', 'z88', 'z89', 'z90', 'z91',
'z92', 'z93', 'z94', 'z95', 'z96', 'z97', 'z98', 'z99', 'z100', 'z101', 'z10
2', 'z103', 'z104', 'z105', 'z106', 'z107', 'z108', 'z109', 'z110', 'z111',
'z112', 'z113', 'z114', 'z115', 'z116', 'z117', 'z118', 'z119', 'z120', 'z12

```
1', 'z122', 'z123', 'z124', 'z125', 'z126', 'z127', 'z128', 'z129', 'z130',
'z131', 'z132', 'z133', 'z134', 'z135', 'z136', 'z137', 'z138', 'z139', 'z14
0', 'z141', 'z142', 'z143', 'z144', 'z145', 'z146', 'z147', 'z148', 'z149',
'z150', 'z151', 'z152', 'z153', 'z154', 'z155', 'z156', 'z157', 'z158', 'z15
9', 'z160', 'z161', 'z162', 'z163', 'z164', 'z165', 'z166', 'z167', 'z168',
'z169', 'z170', 'z171', 'z172', 'z173', 'z174', 'z175', 'z176', 'z177', 'z17
8', 'z179', 'z180', 'z181', 'z182', 'z183', 'z184', 'z185', 'z186', 'z187',
'z188', 'z189', 'z190', 'z191', 'z192', 'z193', 'z194', 'z195', 'z196', 'z19
7', 'z198', 'z199', 'z200', 'z201', 'z202', 'z203', 'z204', 'z205', 'z206',
'z207', 'z208', 'z209', 'z210', 'z211', 'z212', 'z213', 'z214', 'z215', 'z21
6', 'z217', 'z218', 'z219', 'z220', 'z221', 'z222', 'subject', 'phase', 'sta
te', 'output']

Missing values:
x1      0
y219    0
y221    0
y222    0
z1      0
z2      0
z3      0
z4      0
z5      0
z6      0
z7      0
z8      0
z9      0
z10     0
z11     0
z12     0
z13     0
z14     0
z15     0
y220    0
dtype: int64
Unique values:
x1         1
y57        1
y72        1
y73        1
y74        1
         ...
y60     4584
y61     4584
y62     4584
y19     4584
y161    4584
Length: 670, dtype: int64
```

In [14]:

```python
##Checking the class distribution and imbalance in the training data

print("Class distribution:")
print(train['output'].value_counts())

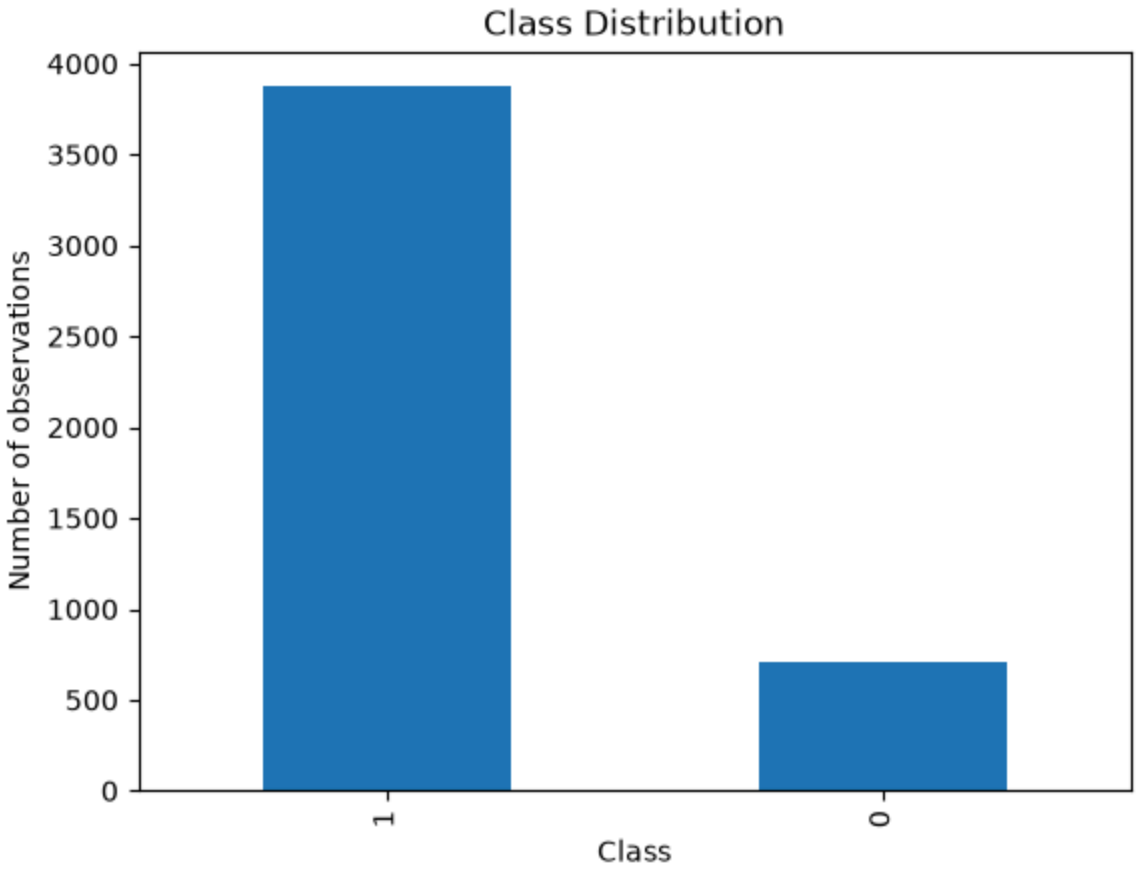
print("\nClass percentages:")
print(train['output'].value_counts(normalize=True) * 100)
```

```
train['output'].value_counts().plot(
    kind='bar',
    title='Class Distribution'
)

plt.xlabel("Class")
plt.ylabel("Number of observations")
plt.show()
```

```
Class distribution:
output
1    3873
0     711
Name: count, dtype: int64

Class percentages:
output
1    84.489529
0    15.510471
Name: proportion, dtype: float64
```



In [15]:
```
# Separate features and target of the training data
X = train.drop(columns=['output'])
y = train['output']

# Stratified train-validation split
X_train, X_val, y_train, y_val = train_test_split(
    X,
    y,
```

```
    test_size=0.20,
    random_state=42,
    stratify=y
)

print("Training observations:", X_train.shape)
print("Validation observations:", X_val.shape)
```

```
Training observations: (3667, 669)
Validation observations: (917, 669)
```

In [16]:
```
# Identify numerical and categorical columns
from sklearn.compose import ColumnTransformer

numeric_features = X.select_dtypes(
    include=['int64', 'float64']
).columns.tolist()

categorical_features = X.select_dtypes(
    include=['object', 'category', 'bool']
).columns.tolist()


# Numeric preprocessing
numeric_transformer = Pipeline([
    ('imputer', SimpleImputer(strategy='median')),
    ('scaler', StandardScaler())
])


# Categorical preprocessing
categorical_transformer = Pipeline([
    ('imputer', SimpleImputer(strategy='most_frequent')),
    ('encoder', OneHotEncoder(handle_unknown='ignore'))
])

# Combine preprocessing
preprocessor = ColumnTransformer([
    ('num', numeric_transformer, numeric_features),
    ('cat', categorical_transformer, categorical_features)
])
```

```
/var/folders/c3/1xdc_j3d5gnbhhnvrtwj5c540000gn/T/ipykernel_2596/3195068650.p
y:8: Pandas4Warning: For backward compatibility, 'str' dtypes are included b
y select_dtypes when 'object' dtype is specified. This behavior is deprecate
d and will be removed in a future version. Explicitly pass 'str' to `include
` to select them, or to `exclude` to remove them and silence this warning.
See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#strin
g-migration-select-dtypes for details on how to write code that works with p
andas 2 and 3.
  categorical_features = X.select_dtypes(
```

In [2]:
```
###### Baseline Model ######

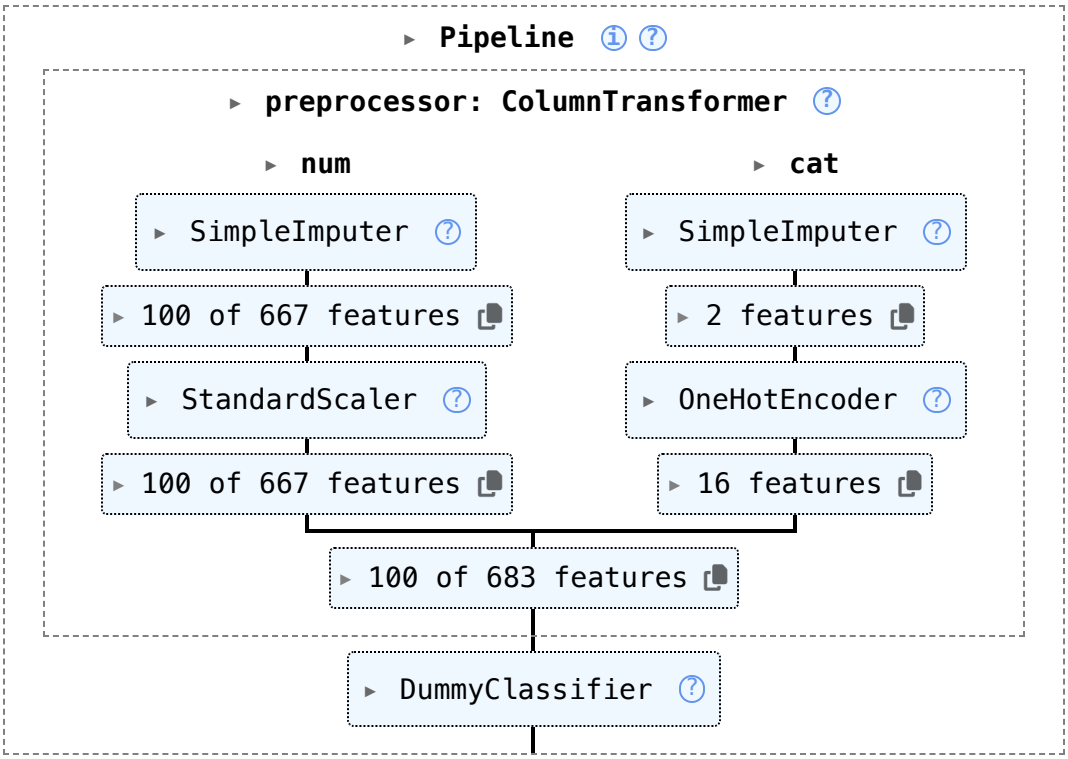
##Before developing sophisticated models, we want to understand the class di
##A majority-class classifier provides the minimum accuracy that any meaning
```

In [ ]:
```python
#Code generate with the help of AI

baseline_model = Pipeline([
    ('preprocessor', preprocessor),
    ('classifier', DummyClassifier(strategy='most_frequent'))
])

baseline_model.fit(X_train, y_train)
```

Out[ ]:



In [ ]:
```python
## Baseline Model Evaluation
#Code generate with the help of AI

# Predictions
baseline_pred = baseline_model.predict(X_val)

# Evaluation
baseline_accuracy = accuracy_score(y_val, baseline_pred)
baseline_precision = precision_score(
    y_val, baseline_pred, zero_division=0
)
baseline_recall = recall_score(
    y_val, baseline_pred, zero_division=0
)
baseline_f1 = f1_score(
    y_val, baseline_pred, zero_division=0
)

print("BASELINE MODEL PERFORMANCE")
print("--------------------------")
print(f"Accuracy : {baseline_accuracy:.4f}")
print(f"Precision: {baseline_precision:.4f}")
print(f"Recall   : {baseline_recall:.4f}")
```

```
print(f"F1 Score : {baseline_f1:.4f}")

print("\nClassification Report:")
print(classification_report(y_val, baseline_pred, zero_division=0))
```

```
BASELINE MODEL PERFORMANCE
--------------------------
Accuracy : 0.8451
Precision: 0.8451
Recall   : 1.0000
F1 Score : 0.9161

Classification Report:
              precision    recall  f1-score   support

           0       0.00      0.00      0.00       142
           1       0.85      1.00      0.92       775

    accuracy                           0.85       917
   macro avg       0.42      0.50      0.46       917
weighted avg       0.71      0.85      0.77       917
```

In [ ]:
```
###### Logistic Regression Model #####

##Next to baseline model, we will implement a logistic regression model.
#Logistic regression is a simple yet effective linear model for binary class
#It estimates the probability of a binary outcome based on one or more predi
#It also tells if the problem is linearlyt separable
```

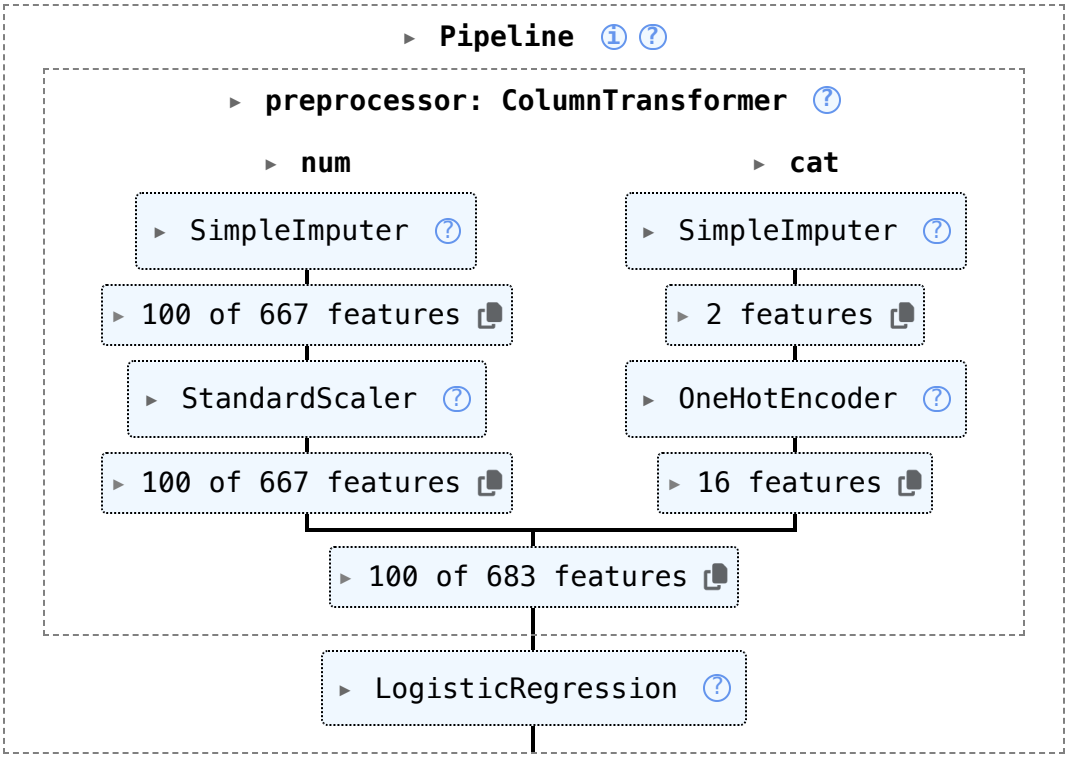
In [14]:
```
##Training a Logistic Regression model with class weights to handle class im

logistic_model = Pipeline([
    ('preprocessor', preprocessor),
    ('classifier', LogisticRegression(
        max_iter=2000,
        class_weight='balanced',
        solver='lbfgs',
        C=1.0,
        random_state=42
    ))
])
```

In [15]:
```
# Fit the logistic regression model

logistic_model.fit(X_train, y_train)
```

Out[15]:



In [ ]:

```
## Model Evaluation for Logistic Regression
#Code generate with the help of AI

# Predictions for Logistic Regression model
logistic_pred = logistic_model.predict(X_val)

# Predicted probabilities for positive class
logistic_prob = logistic_model.predict_proba(X_val)[:, 1]

# Metrics
logistic_accuracy = accuracy_score(y_val, logistic_pred)

logistic_precision = precision_score(
    y_val, logistic_pred, zero_division=0
)

logistic_recall = recall_score(
    y_val, logistic_pred, zero_division=0
)

logistic_f1 = f1_score(
    y_val, logistic_pred, zero_division=0
)

logistic_auc = roc_auc_score(y_val, logistic_prob)

print("LOGISTIC REGRESSION PERFORMANCE")
print("-------------------------------")
print(f"Accuracy : {logistic_accuracy:.4f}")
print(f"Precision: {logistic_precision:.4f}")
print(f"Recall   : {logistic_recall:.4f}")
print(f"F1 Score : {logistic_f1:.4f}")
print(f"ROC-AUC  : {logistic_auc:.4f}")
```

```
print("\nClassification Report:")
print(classification_report(y_val, logistic_pred))
```

```
LOGISTIC REGRESSION PERFORMANCE
--------------------------------
Accuracy : 0.7328
Precision: 0.9288
Recall   : 0.7406
F1 Score : 0.8241
ROC-AUC  : 0.7908

Classification Report:
              precision    recall  f1-score   support

           0       0.33      0.69      0.44       142
           1       0.93      0.74      0.82       775

    accuracy                           0.73       917
   macro avg       0.63      0.72      0.63       917
weighted avg       0.84      0.73      0.77       917
```

In [17]:
```
## Model Comparison between Baseline and Logistic Regression

comparison = pd.DataFrame({
    'Model': [
        'Baseline',
        'Logistic Regression'
    ],
    'Accuracy': [
        baseline_accuracy,
        logistic_accuracy
    ],
    'Precision': [
        baseline_precision,
        logistic_precision
    ],
    'Recall': [
        baseline_recall,
        logistic_recall
    ],
    'F1 Score': [
        baseline_f1,
        logistic_f1
    ]
})

display(comparison)
```

	Model	Accuracy	Precision	Recall	F1 Score
0	Baseline	0.845147	0.845147	1.000000	0.916076
1	Logistic Regression	0.732824	0.928803	0.740645	0.824121

In [ ]:
```
#Train Logistic Regression on the entire training dataset
#Code generated with the help of AI

X_full = train.drop(columns=['output'])
y_full = train['output']

logistic_model.fit(X_full, y_full)

# Predict probabilities for the Kaggle test dataset
preds = logistic_model.predict_proba(test)

# Create submission DataFrame
submithis = pd.DataFrame([
    test.index,
    preds[:, 1]
]).T

submithis.columns = ['id', 'output']
submithis.id = submithis.id.astype(int)

display(submithis.head())

# Save submission
submithis.to_csv('submission_logistic.csv', index=False)

print("Submission file created successfully!")
```

	id	output
0	0	0.161697
1	1	0.960852
2	2	0.536888
3	3	0.661943
4	4	0.675553

```
Submission file created successfully!
```

In [19]:
```
##Generate learning curves for the logistic regression model
from sklearn.model_selection import StratifiedKFold

# Stratified cross-validation
cv = StratifiedKFold(
    n_splits=5,
    shuffle=True,
    random_state=42
)

# Training dataset sizes
train_sizes = np.linspace(0.1, 1.0, 8)

# Calculate learning curves
train_sizes_abs, train_scores, val_scores = learning_curve(
    estimator=logistic_model,
```

```
    X=X_train,
    y=y_train,
    train_sizes=train_sizes,
    cv=cv,
    scoring='f1',
    n_jobs=-1,
    shuffle=True,
    random_state=42
)

# Mean scores
train_mean = np.mean(train_scores, axis=1)
val_mean = np.mean(val_scores, axis=1)

# Standard deviations
train_std = np.std(train_scores, axis=1)
val_std = np.std(val_scores, axis=1)
```

In [20]:

```
##Generate learning curve plot

plt.figure(figsize=(10, 6))

# Training curve
plt.plot(
    train_sizes_abs,
    train_mean,
    marker='o',
    label='Training F1 Score'
)

# Validation curve
plt.plot(
    train_sizes_abs,
    val_mean,
    marker='s',
    label='Cross-Validation F1 Score'
)

# Training variance
plt.fill_between(
    train_sizes_abs,
    train_mean - train_std,
    train_mean + train_std,
    alpha=0.15
)

# Validation variance
plt.fill_between(
    train_sizes_abs,
    val_mean - val_std,
    val_mean + val_std,
    alpha=0.15
)

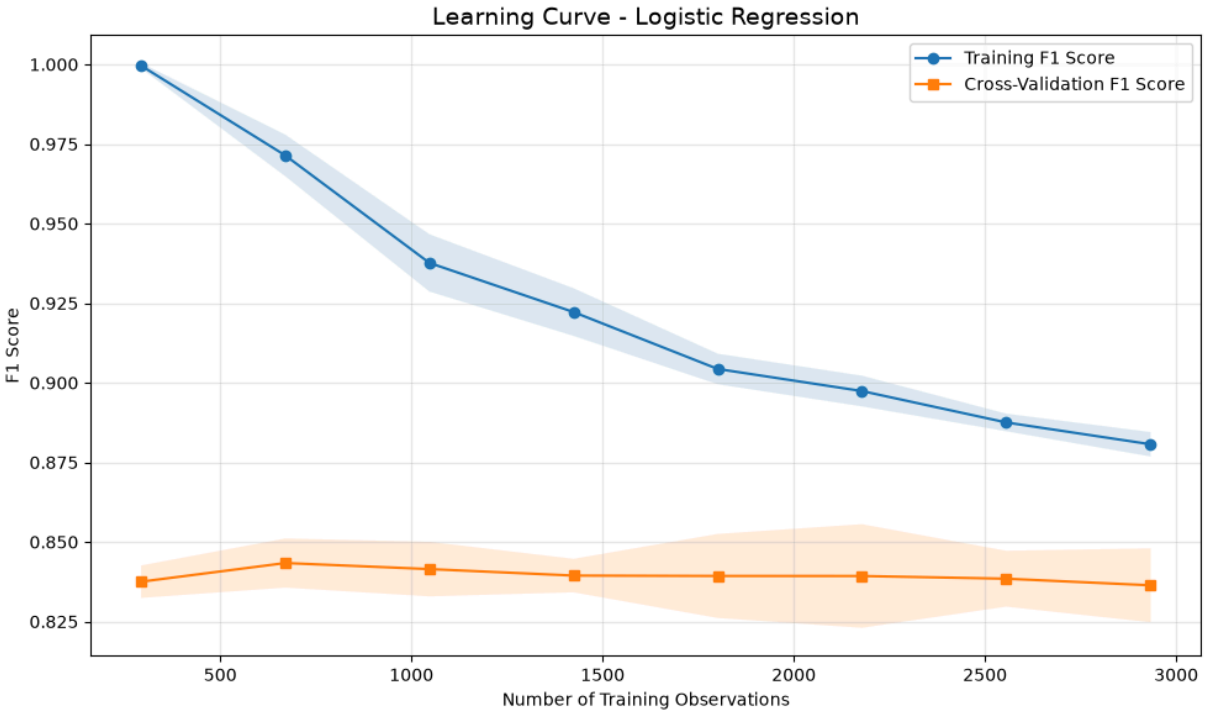
plt.title(
    'Learning Curve - Logistic Regression',
```

```
    fontsize=14
)

plt.xlabel('Number of Training Observations')
plt.ylabel('F1 Score')

plt.legend()
plt.grid(alpha=0.3)
plt.tight_layout()

plt.show()
```



In [ ]:

```
#Code generated with the help of AI

def diagnose_model(
    train_scores,
    val_scores,
    gap_threshold=0.10,
    low_score_threshold=0.60
):

    train_f1 = np.mean(train_scores[-3:])
    val_f1 = np.mean(val_scores[-3:])

    gap = train_f1 - val_f1

    print("MODEL DIAGNOSTICS")
    print("-----------------------------")
    print(f"Training F1   : {train_f1:.4f}")
    print(f"Validation F1 : {val_f1:.4f}")
    print(f"Gap           : {gap:.4f}")

    if gap > gap_threshold:
```

```
        print("\nDiagnosis: Possible OVERFITTING")
        print("Training performance is considerably higher.")
        print("Consider stronger regularization or more data.")

    elif (
        train_f1 < low_score_threshold
        and val_f1 < low_score_threshold
    ):

        print("\nDiagnosis: Possible UNDERFITTING")
        print("Both training and validation scores are low.")
        print("Consider feature engineering or a more flexible model.")

    else:

        print("\nDiagnosis: No obvious severe overfitting.")
        print("Assess performance against your baseline.")
```

In [22]:
```
diagnose_model(
    train_mean,
    val_mean
)
```

```
MODEL DIAGNOSTICS
------------------------------
Training F1   : 0.8885
Validation F1 : 0.8380
Gap           : 0.0505

Diagnosis: No obvious severe overfitting.
Assess performance against your baseline.
```

Improvement: Feature Selection using Logistic Regression with L1 regularization

In [ ]:
```
#Improvement: Feature Selection using Logistic Regression with L1 regulariza
#This improvement aims to enhance the model's performance by selecting the m

from sklearn.feature_selection import SelectFromModel

feature_selection_lr = Pipeline([
    ('preprocessor', preprocessor),

    ('feature_selection', SelectFromModel(
        LogisticRegression(
            penalty='l1',
            solver='liblinear',
            C=0.1,
            max_iter=3000
        )
    )),

    ('classifier', LogisticRegression(
        C=1.0,
        max_iter=3000,
        class_weight='balanced',
```

```
        random_state=42
    ))
])

feature_selection_lr.fit(X_train, y_train)

preds = feature_selection_lr.predict_proba(X_val)[:, 1]

print(
    "Feature Selection ROC-AUC:",
    roc_auc_score(y_val, preds)
)
```

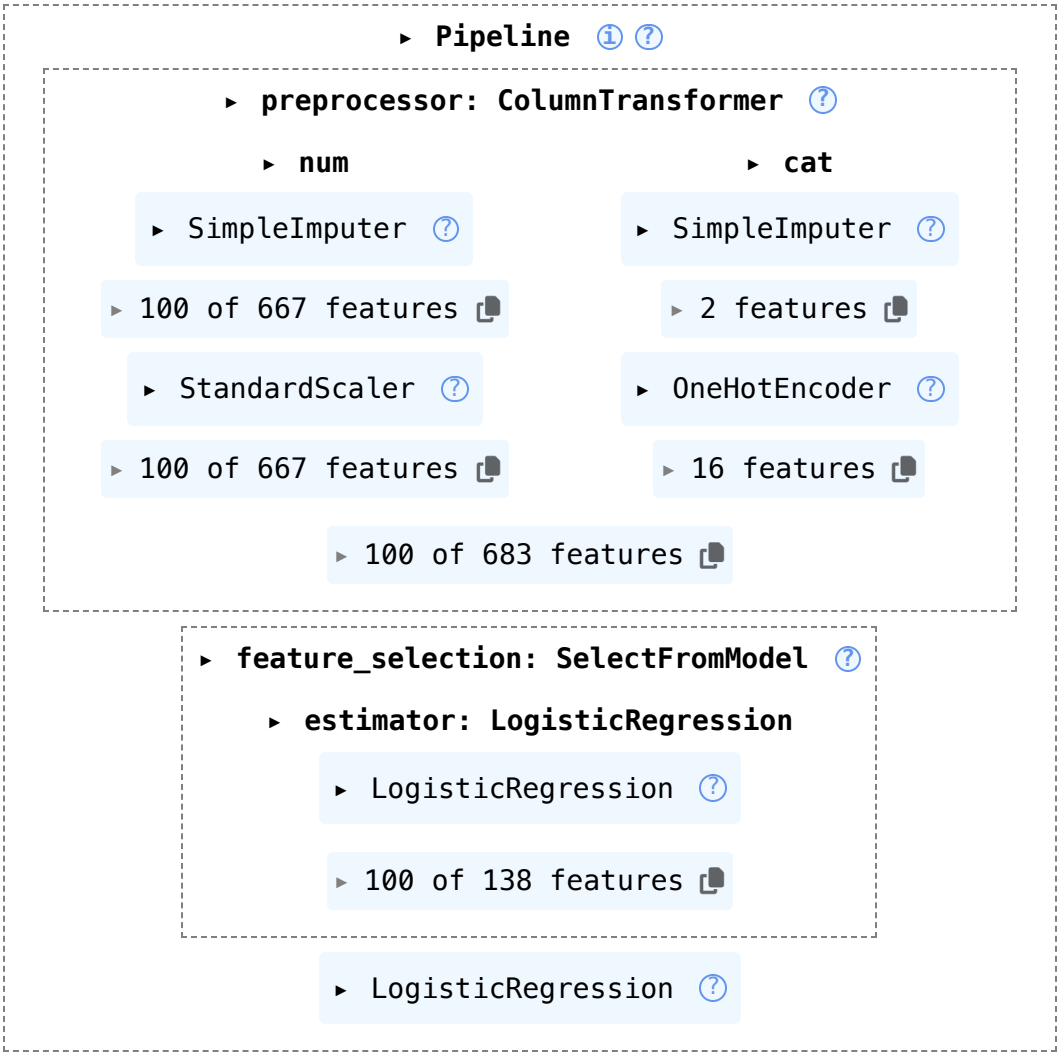
```
/opt/miniconda3/envs/section_5/lib/python3.14/site-packages/sklearn/linear_model/_logistic.py:1381: FutureWarning: 'penalty' was deprecated in version 1.8 and will be removed in 1.10. To avoid this warning, leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', l1_ratio set to a float between 0 and 1 instead of penalty='elasticnet', and C=np.inf instead of penalty=None.
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  warnings.warn(
Feature Selection ROC-AUC: 0.8124307133121309
```

In [14]:
```
# Train Logistic Regression with Feature Selection
# using the complete labeled training dataset

feature_selection_lr.fit(X, y)
```

```
/opt/miniconda3/envs/section_5/lib/python3.14/site-packages/sklearn/linear_model/_logistic.py:1381: FutureWarning: 'penalty' was deprecated in version 1.8 and will be removed in 1.10. To avoid this warning, leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', l1_ratio set to a float between 0 and 1 instead of penalty='elasticnet', and C=np.inf instead of penalty=None.
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  warnings.warn(
```

Out[14]:



In [ ]:

```
#Code generated with the help of AI

# Prepare test features
test_1 = test.reindex(columns=X.columns)

# Generate predicted probabilities
preds = feature_selection_lr.predict_proba(test_1)

# Create submission DataFrame
submithis = pd.DataFrame([test_1.index, preds[:, 1]]).T
submithis.columns = ['id', 'output']
submithis.id = submithis.id.astype(int)

# Display submission
submithis
```

Out[ ]:

	id	output
0	0	0.359823
1	1	0.840429
2	2	0.520054
3	3	0.520288
4	4	0.746260
...	...	...
1727	1727	0.700496
1728	1728	0.618997
1729	1729	0.567410
1730	1730	0.804203
1731	1731	0.784516

1732 rows × 2 columns

In [ ]:
```
# Save the submission file for the Logistic Regression model with Feature Se

submithis.to_csv('submission_feature_selection.csv', index=False)
```

In [42]:
```
##Improved Logistic Regression with Hyperparameter Optimization
```

In [17]:
```
##Improved Logistic Regression Pipeline with Hyperparameter Optimization


##Improvements:
##L1 feature selection to remove less useful features.
##Elastic Net regularization to balance feature sparsity and coefficient sta
##Hyperparameter tuning using stratified cross-validation.
##ROC-AUC optimization rather than accuracy.
##Tests different class-weight configurations.
##Automatically selects the best combination of parameters.

from sklearn.pipeline import Pipeline
from sklearn.linear_model import LogisticRegression
from sklearn.feature_selection import SelectFromModel
from sklearn.model_selection import RandomizedSearchCV, StratifiedKFold
from sklearn.metrics import roc_auc_score
from scipy.stats import loguniform

# Faster Logistic Regression
fast_lr = Pipeline([

    ('preprocessor', preprocessor),

    ('feature_selection', SelectFromModel(
```

```
        LogisticRegression(
            penalty='l1',
            solver='liblinear',
            C=0.1,
            max_iter=1000,
            random_state=42
        )
    )),

    ('classifier', LogisticRegression(
        penalty='l2',
        solver='liblinear',
        max_iter=1000,
        random_state=42
    ))
])

# Smaller hyperparameter space
param_dist = {

    'feature_selection__estimator__C': [
        0.01, 0.1, 1.0
    ],

    'classifier__C': [
        0.01, 0.1, 1.0, 10
    ],

    'classifier__class_weight': [
        None, 'balanced'
    ]
}

# 3-fold cross-validation
cv = StratifiedKFold(
    n_splits=3,
    shuffle=True,
    random_state=42
)

# Faster search
lr_search = RandomizedSearchCV(
    estimator=fast_lr,
    param_distributions=param_dist,
    n_iter=10,
    scoring='roc_auc',
    cv=cv,
    n_jobs=2,
    verbose=1,
    random_state=42
)

# Train
lr_search.fit(X_train, y_train)

# Best model
```

```python
best_lr = lr_search.best_estimator_

# Predictions
lr_probs = best_lr.predict_proba(X_val)[:, 1]

print("Best Parameters:", lr_search.best_params_)

print("Best CV ROC-AUC:", lr_search.best_score_)

print(
    "Validation ROC-AUC:",
    roc_auc_score(y_val, lr_probs)
)
```

```
Fitting 3 folds for each of 10 candidates, totalling 30 fits
```

```
/opt/miniconda3/envs/section_5/lib/python3.14/site-packages/sklearn/linear_m
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/opt/miniconda3/envs/section_5/lib/python3.14/site-packages/sklearn/linear_model/_logistic.py:1381: FutureWarning: 'penalty' was deprecated in version 1.8 and will be removed in 1.10. To avoid this warning, leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', l1_ratio set to a float between 0 and 1 instead of penalty='elasticnet', and C=np.inf instead of penalty=None.
  warnings.warn(
/opt/miniconda3/envs/section_5/lib/python3.14/site-packages/sklearn/linear_model/_logistic.py:1381: FutureWarning: 'penalty' was deprecated in version 1.8 and will be removed in 1.10. To avoid this warning, leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', l1_ratio set to a float between 0 and 1 instead of penalty='elasticnet', and C=np.inf instead of penalty=None.
  warnings.warn(
/opt/miniconda3/envs/section_5/lib/python3.14/site-packages/sklearn/linear_model/_logistic.py:1407: UserWarning: Inconsistent values: penalty=l1 with l1_ratio=0.0. penalty is deprecated. Please use l1_ratio only.
  warnings.warn(
Best Parameters: {'feature_selection__estimator__C': 0.1, 'classifier__class_weight': 'balanced', 'classifier__C': 1.0}
Best CV ROC-AUC: 0.7904314889238165
Validation ROC-AUC: 0.8124034529759201
```

```
/opt/miniconda3/envs/section_5/lib/python3.14/site-packages/sklearn/linear_model/_logistic.py:1381: FutureWarning: 'penalty' was deprecated in version 1.8 and will be removed in 1.10. To avoid this warning, leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', l1_ratio set to a float between 0 and 1 instead of penalty='elasticnet', and C=np.inf instead of penalty=None.
  warnings.warn(
```

In [18]:
```python
##Evaluate how many features were selected by the feature selection step in

# Retrieve fitted feature selector
selector = best_lr.named_steps['feature_selection']

# Retrieve feature names after preprocessing
feature_names = best_lr.named_steps[
    'preprocessor'
].get_feature_names_out()

# Identify selected features
selected_mask = selector.get_support()

selected_features = feature_names[selected_mask]

print("Total Features:", len(feature_names))

print("Selected Features:", len(selected_features))

print("Removed Features:", len(feature_names) - len(selected_features))
```

```
Total Features: 683
Selected Features: 118
Removed Features: 565
```

In [19]:
```python
##Visualize the top 20 features based on their coefficients in the improved

import pandas as pd
import matplotlib.pyplot as plt

# Extract coefficients
coefficients = best_lr.named_steps[
    'classifier'
].coef_[0]

# Create importance dataframe
importance_df = pd.DataFrame({
    'Feature': selected_features,
    'Coefficient': coefficients,
    'Importance': np.abs(coefficients)
})

# Sort by absolute importance
importance_df = importance_df.sort_values(
    by='Importance',
    ascending=False
)
```

```python
print(importance_df.head(20))

# Plot top 20 features
plt.figure(figsize=(10, 8))

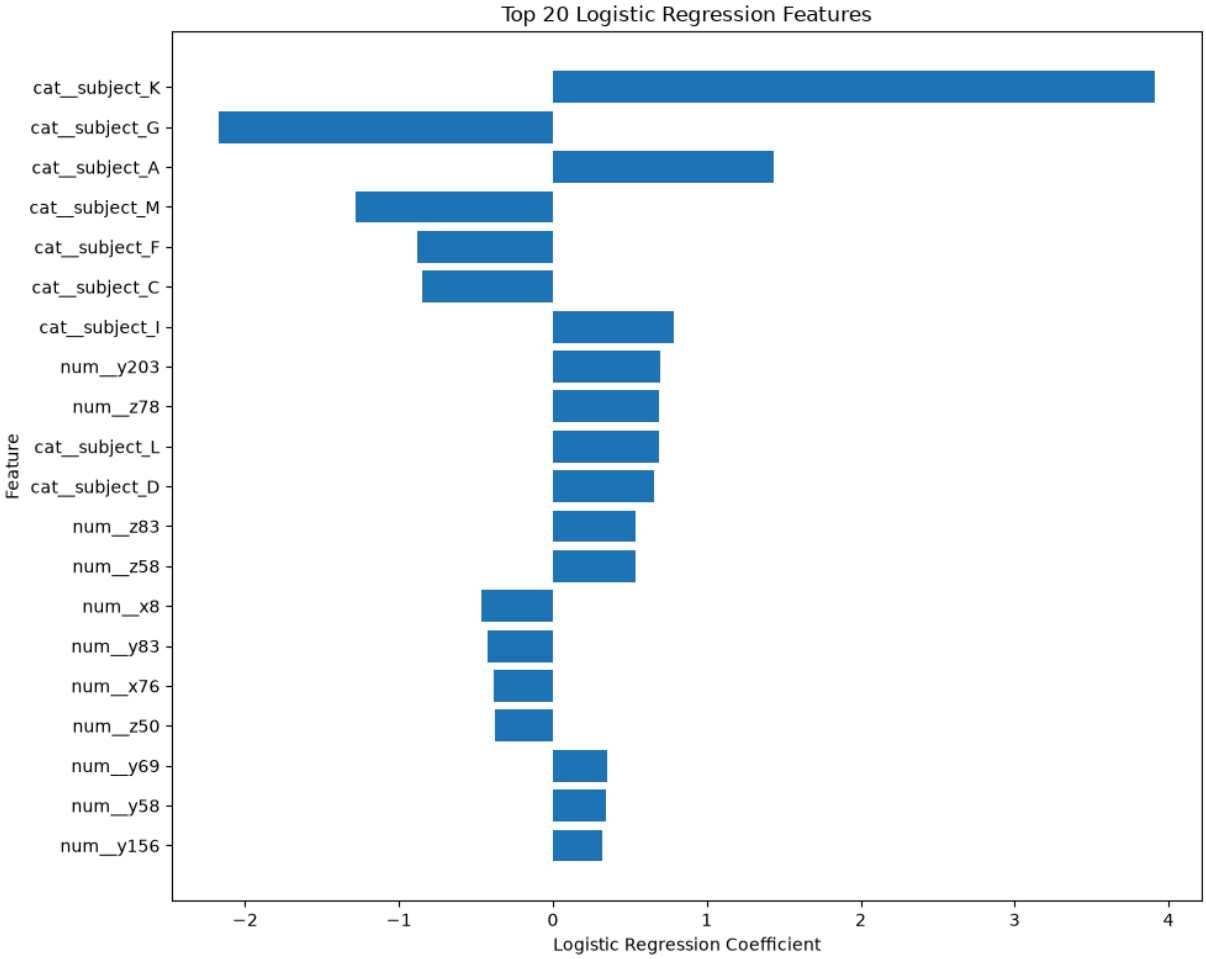
top_features = importance_df.head(20).iloc[::-1]

plt.barh(
    top_features['Feature'],
    top_features['Coefficient']
)

plt.xlabel("Logistic Regression Coefficient")
plt.ylabel("Feature")
plt.title("Top 20 Logistic Regression Features")

plt.tight_layout()
plt.show()
```

```
           Feature  Coefficient  Importance
114  cat__subject_K     3.911539    3.911539
112  cat__subject_G    -2.170483    2.170483
108  cat__subject_A     1.432577    1.432577
116  cat__subject_M    -1.285749    1.285749
111  cat__subject_F    -0.882752    0.882752
109  cat__subject_C    -0.851302    0.851302
113  cat__subject_I     0.786729    0.786729
61        num__y203     0.694713    0.694713
86         num__z78     0.691326    0.691326
115  cat__subject_L     0.686760    0.686760
110  cat__subject_D     0.655168    0.655168
87         num__z83     0.539739    0.539739
80         num__z58     0.533655    0.533655
2           num__x8    -0.460499    0.460499
46         num__y83    -0.420472    0.420472
10         num__x76    -0.383717    0.383717
78         num__z50    -0.379353    0.379353
43         num__y69     0.356056    0.356056
41         num__y58     0.341020    0.341020
57        num__y156     0.316906    0.316906
```



In [ ]:

```python
# Load sample solution
#Code generated with the help of AI

sample_solution = pd.read_csv(
    "sample_solution.csv"
)

# Prepare test data
test_1 = test.reindex(columns=X_train.columns)

# Generate predictions
test_probs = best_lr.predict_proba(test_1)[:, 1]

# Create submission
submission = sample_solution.copy()

submission["output"] = test_probs

# Save submission file
submission.to_csv(
    "logistic_regression_optimized_submission.csv",
    index=False
)

print(submission.head())
```

```
print("Submission shape:", submission.shape)
```

```
   id    output
0   0  0.327589
1   1  0.711085
2   2  0.564783
3   3  0.511049
4   4  0.745510
Submission shape: (1732, 2)
```

In [18]:
```
##### XGBoost Classifier ########

###XGBoost is a powerful gradient boosting algorithm that is widely used for
##This model is particularly effective for handling complex relationships ar
```

We use XGBoost because the dataset contains many features, nonlinear relationships, missing values, and substantial class imbalance, XGBoost may outperform simpler models. Gradient boosting allows each successive tree to focus on errors from previous trees, while regularization, feature subsampling, row subsampling, and class weighting can help reduce overfitting and address imbalance. XGBoost also provides feature importance and integrates well with SHAP, making it suitable for both predictive performance and model explainability.

In [17]:
```
#Code generated with the help of AI

from xgboost import XGBClassifier

# Calculate class imbalance ratio
class_counts = y_train.value_counts()

scale_pos_weight = (
    class_counts[0] / class_counts[1]
)

print("Scale Positive Weight:", scale_pos_weight)

xgb_model = Pipeline([
    ('preprocessor', preprocessor),

    ('classifier', XGBClassifier(
        n_estimators=500,
        max_depth=4,
        learning_rate=0.03,
        subsample=0.8,
        colsample_bytree=0.8,
        objective='binary:logistic',
        eval_metric='auc',
        random_state=42,
        n_jobs=2
    ))
])
```

```
xgb_model.fit(X_train, y_train)

xgb_preds = xgb_model.predict_proba(X_val)[:, 1]

print(
    "XGBoost Validation ROC-AUC:",
    roc_auc_score(y_val, xgb_preds)
)
```

```
Scale Positive Weight: 0.18366688185926405
XGBoost Validation ROC-AUC: 0.859418446160836
```

In [ ]:
```
# Prepare test data
test_1 = test.reindex(columns=X.columns)

# Generate predicted probabilities
preds = xgb_model.predict_proba(test_1)

# Create submission DataFrame
submithis = pd.DataFrame([test_1.index, preds[:, 1]]).T

submithis.columns = ['id', 'output']

submithis.id = submithis.id.astype(int)

# Display submission
submithis
```

Out[ ]:

	id	output
0	0	0.890614
1	1	0.954558
2	2	0.932373
3	3	0.762751
4	4	0.834526
...	...	...
1727	1727	0.932693
1728	1728	0.949144
1729	1729	0.936582
1730	1730	0.940488
1731	1731	0.954646

1732 rows × 2 columns

In [23]:
```
# Save submission file
submithis.to_csv(
    'submission_xgboost.csv',
```

```
    index=False
)
```

In [44]:
```
######Improved XGBoost Classifier with Hyperparameter Optimization, early st
```

In [ ]:
```
#Code generated with the help of AI

import numpy as np
import pandas as pd

from sklearn.base import clone
from sklearn.metrics import roc_auc_score
from xgboost import XGBClassifier


# Calculate class imbalance
class_counts = y_train.value_counts()

scale_pos_weight = (
    class_counts[0] / class_counts[1]
)

print("Class distribution:")
print(class_counts)

print("Scale Positive Weight:", scale_pos_weight)



# Preprocess training and validation data
optimized_preprocessor = clone(preprocessor)

X_train_processed = optimized_preprocessor.fit_transform(
    X_train
)

X_val_processed = optimized_preprocessor.transform(
    X_val
)


# Build optimized XGBoost
optimized_xgb = XGBClassifier(

    n_estimators=2000,
    max_depth=3,
    learning_rate=0.025,
    min_child_weight=5,
    subsample=0.85,
    colsample_bytree=0.75,
    gamma=0.1,
    reg_alpha=0.5,
    reg_lambda=5,
    scale_pos_weight=1,
```

```
    objective='binary:logistic',
    eval_metric='auc',
    tree_method='hist',
    early_stopping_rounds=75,
    random_state=42,
    n_jobs=2

)

# Train with early stopping
optimized_xgb.fit(

    X_train_processed,
    y_train,
    eval_set=[
        (X_val_processed, y_val)
    ],

    verbose=False
)

# Validate
optimized_preds = optimized_xgb.predict_proba(
    X_val_processed
)[:, 1]

optimized_auc = roc_auc_score(
    y_val,
    optimized_preds
)

print(
    "Optimized XGBoost ROC-AUC:",
    optimized_auc
)

print(
    "Best iteration:",
    optimized_xgb.best_iteration
)
```

```
Class distribution:
output
1    3098
0     569
Name: count, dtype: int64
Scale Positive Weight: 0.18366688185926405
Optimized XGBoost ROC-AUC: 0.8509859154929579
Best iteration: 311
```

In [ ]:

```
### Feature Selection using the optimized XGBoost model

from sklearn.feature_selection import SelectFromModel
from sklearn.metrics import roc_auc_score

# Feature selection
```

```
selector = SelectFromModel(

    optimized_xgb,
    prefit=True,
    threshold=-np.inf,
    max_features=150
)

X_train_selected = selector.transform(
    X_train_processed
)

X_val_selected = selector.transform(
    X_val_processed
)

print(
    "Original number of processed features:",
    X_train_processed.shape[1]
)

print(
    "Selected features:",
    X_train_selected.shape[1]
)


# Train new XGBoost with selected features
selected_xgb = XGBClassifier(

    n_estimators=2000,
    max_depth=3,
    learning_rate=0.025,
    min_child_weight=5,
    subsample=0.85,
    colsample_bytree=0.75,
    gamma=0.1,
    reg_alpha=0.5,
    reg_lambda=5,
    scale_pos_weight=1,
    objective='binary:logistic',
    eval_metric='auc',
    tree_method='hist',
    early_stopping_rounds=75,
    random_state=42,
    n_jobs=2
)


selected_xgb.fit(

    X_train_selected,
    y_train,

    eval_set=[
        (X_val_selected, y_val)
```

```
    ],
    verbose=False
)


# Evaluate

selected_preds = selected_xgb.predict_proba(
    X_val_selected
)[:, 1]

selected_auc = roc_auc_score(
    y_val,
    selected_preds
)

print(
    "Feature Selected XGBoost ROC-AUC:",
    selected_auc
)
```

```
Original number of processed features: 683
Selected features: 150
Feature Selected XGBoost ROC-AUC: 0.8589186733303044
```

In [ ]:

```
### Compare all three models: Original XGBoost, Optimized XGBoost, and Featu
#Code generated with the help of AI

results = pd.DataFrame({

    "Model": [
        "Original XGBoost",
        "Optimized XGBoost",
        "Feature Selected XGBoost"
    ],

    "Validation ROC-AUC": [
        roc_auc_score(y_val, xgb_preds),
        optimized_auc,
        selected_auc
    ]

})

results = results.sort_values(
    by="Validation ROC-AUC",
    ascending=False
)

print(results)
```

```
                     Model  Validation ROC-AUC
0          Original XGBoost            0.866215
2  Feature Selected XGBoost            0.858919
1         Optimized XGBoost            0.850986
```

In [ ]:

```
# Select best model
#Code generated with the help of AI

if selected_auc > optimized_auc:
    best_model = selected_xgb
    print("Using Feature Selected XGBoost")

else:
    best_model = optimized_xgb
    print("Using Optimized XGBoost")


# Prepare test data

test_1 = test.reindex(columns=X.columns)
test_processed = optimized_preprocessor.transform(
    test_1
)


# Apply feature selection if necessary

if best_model is selected_xgb:

    test_processed = selector.transform(
        test_processed
    )


# Generate probabilities
test_probs = best_model.predict_proba(
    test_processed
)[:, 1]


# Load sample solution
sample_solution = pd.read_csv(

    "sample_solution.csv"

)

# Create submission
submission = sample_solution.copy()
submission["output"] = test_probs


# Save submission

submission.to_csv(
    "xgboost_optimized_submission.csv",
    index=False
)
```

```
# Verify
print(submission.head())

print(
    "Submission shape:",
    submission.shape
)

print(
    "Missing predictions:",
    submission["output"].isnull().sum()
)
```

```
Using Feature Selected XGBoost
   id    output
0   0  0.895186
1   1  0.950224
2   2  0.870076
3   3  0.731898
4   4  0.893090
Submission shape: (1732, 2)
Missing predictions: 0
```

In [ ]:
```
##### CatBoost Classifier #####

##We use catboost, a gradient boosting library that handles categorical feat
##This model is particularly effective for datasets with categorical variabl
```

In [19]:
```
import pandas as pd
import numpy as np

from catboost import CatBoostClassifier
from sklearn.metrics import roc_auc_score

#Code generated with the help of AI

# Prepare training and validation data
X_train_cat = X_train.copy()
X_val_cat = X_val.copy()

# Identify categorical columns
cat_features = X_train_cat.select_dtypes(
    include=["object", "category", "string"]
).columns.tolist()

# Convert categorical missing values to strings
for col in cat_features:

    X_train_cat[col] = (
        X_train_cat[col].fillna("Missing").astype(str)
    )

    X_val_cat[col] = (
        X_val_cat[col].fillna("Missing").astype(str)
    )
```

```
# Initialize CatBoost
cat_model = CatBoostClassifier(

    iterations=2000,
    depth=5,
    learning_rate=0.025,
    l2_leaf_reg=5,
    loss_function="Logloss",
    eval_metric="AUC",
    random_seed=42,
    early_stopping_rounds=100,
    verbose=False

)

# Train
cat_model.fit(

    X_train_cat,
    y_train,
    cat_features=cat_features,
    eval_set=(X_val_cat, y_val)

)

# Predictions
cat_preds = cat_model.predict_proba(
    X_val_cat
)[:, 1]

# Evaluate
cat_auc = roc_auc_score(
    y_val,
    cat_preds
)

print("CatBoost Validation ROC-AUC:", cat_auc)

print("Best iteration:", cat_model.best_iteration_)
```

```
CatBoost Validation ROC-AUC: 0.8673512039981827
Best iteration: 441
```

In [20]:

```
# Prepare test data
#Code generated with the help of AI

test_cat = test.reindex(
    columns=X_train_cat.columns
).copy()

for col in cat_features:

    test_cat[col] = (
        test_cat[col].fillna("Missing").astype(str)
    )
```

```
# Generate predictions
cat_test_probs = cat_model.predict_proba(
    test_cat
)[:, 1]

# Load sample solution
sample_solution = pd.read_csv(
    "sample_solution.csv"
)

# Create submission
submission = sample_solution.copy()
submission["output"] = cat_test_probs

submission.to_csv(
    "catboost_submission.csv",
    index=False
)

print(submission.head())
print("Submission shape:", submission.shape)
```

```
   id    output
0   0  0.613334
1   1  0.951838
2   2  0.832086
3   3  0.785167
4   4  0.616044
Submission shape: (1732, 2)
```

In [ ]:
```
##### LightGBM Classifier #####

##LightGBM – particularly strong for large datasets with many rows/features.
##This model is designed to be efficient and scalable, making it suitable fo
```

In [ ]:
```
import lightgbm as lgb

from sklearn.pipeline import Pipeline
from sklearn.metrics import roc_auc_score

#Code generated with the help of AI

lgb_model = Pipeline([

    ("preprocessor", preprocessor),
    ("classifier", lgb.LGBMClassifier(

        n_estimators=700,
        learning_rate=0.025,
        num_leaves=15,
        max_depth=5,
        min_child_samples=30,
        subsample=0.8,
        subsample_freq=1,
```

```
        colsample_bytree=0.75,
        reg_alpha=0.5,
        reg_lambda=5,
        objective="binary",
        verbosity=-1,
        random_state=42,
        n_jobs=2

    ))

])

lgb_model.fit(
    X_train,
    y_train
)

lgb_preds = lgb_model.predict_proba(
    X_val
)[:, 1]

lgb_auc = roc_auc_score(
    y_val,
    lgb_preds
)

print(
    "LightGBM Validation ROC-AUC:",
    lgb_auc
)
```

```
LightGBM Validation ROC-AUC: 0.8585006815084053
```

In [ ]:

```
# Generate probabilities and submission file

test_1 = test.reindex(columns=X.columns)

lgb_test_probs = lgb_model.predict_proba(
    test_1
)[:, 1]

sample_solution = pd.read_csv(
    "sample_solution.csv"
)

submission = sample_solution.copy()

submission["output"] = lgb_test_probs

submission.to_csv(
    "lightgbm_submission.csv",
    index=False
)

print(submission.head())
```

```
   id    output
0   0  0.874888
1   1  0.964974
2   2  0.927976
3   3  0.793542
4   4  0.834734
```

In [ ]:
```
#### Ensemble Classifier using XGBoost, CatBoost, and LightGBM ####

#This is an ensemble model that combines the predictions of XGBoost, CatBoos
#The ensemble approach can often yield better performance by leveraging the
```

In [ ]:
```
#Code generated with the help of AI

from sklearn.metrics import roc_auc_score

# Ensemble probabilities
ensemble_preds = (

    0.40 * cat_preds +
    0.35 * optimized_preds +
    0.25 * lgb_preds

)

ensemble_auc = roc_auc_score(
    y_val,
    ensemble_preds
)

print(
    "Ensemble Validation ROC-AUC:",
    ensemble_auc
)
```

```
Ensemble Validation ROC-AUC: 0.8655247614720583
```

In [ ]:
```
# Prepare test dataset
test_1 = test.reindex(columns=X.columns)

# XGBoost probabilities
test_processed = optimized_preprocessor.transform(
    test_1
)

xgb_test_probs = optimized_xgb.predict_proba(
    test_processed
)[:, 1]

# CatBoost probabilities
cat_test_probs = cat_model.predict_proba(
    test_cat
)[:, 1]

# LightGBM probabilities
```

```
lgb_test_probs = lgb_model.predict_proba(
    test_1
)[:, 1]

# Ensemble
ensemble_test_probs = (

    0.40 * cat_test_probs +
    0.35 * xgb_test_probs +
    0.25 * lgb_test_probs

)

# ensemblesubmission
sample_solution = pd.read_csv(
    "sample_solution.csv"
)

submission = sample_solution.copy()

submission["output"] = ensemble_test_probs

submission.to_csv(
    "ensemble_submission.csv",
    index=False
)

print(submission.head())

print("Submission shape:", submission.shape)
```

```
   id    output
0   0  0.780004
1   1  0.945674
2   2  0.860053
3   3  0.767698
4   4  0.765722
Submission shape: (1732, 2)
```

In [50]:
```
#### Naive Bayes Classifier ####

##Gaussian Naive Bayes is a simple probabilistic classifier based on applyin
##Optimized Naive Bayes Classifier with hyperparameter tuning and feature se
##Naive Bayes with feature selection and hyperparameter tuning
```

In [ ]:
```
#Code generated with the help of AI

import pandas as pd
import numpy as np

from sklearn.naive_bayes import GaussianNB

from sklearn.pipeline import Pipeline
from sklearn.preprocessing import FunctionTransformer
from sklearn.metrics import (
```

```
    accuracy_score,
    roc_auc_score,
    classification_report,
    confusion_matrix
)

# Create Gaussian Naive Bayes pipeline
nb_model = Pipeline([
    ('preprocessor', preprocessor),

    # Convert sparse matrices to dense if necessary
    ('to_dense', FunctionTransformer(
        lambda x: x.toarray() if hasattr(x, 'toarray') else x
    )),

    ('classifier', GaussianNB())
])

# Train model
nb_model.fit(X_train, y_train)

# Predict classes
nb_predictions = nb_model.predict(X_val)

# Predict probabilities
nb_probs = nb_model.predict_proba(X_val)[:, 1]

# Evaluate
print("Gaussian Naive Bayes Results")
print("----------------------------")

print(
    "Accuracy:",
    accuracy_score(y_val, nb_predictions)
)

print(
    "ROC-AUC:",
    roc_auc_score(y_val, nb_probs)
)

print("\nClassification Report:")
print(classification_report(y_val, nb_predictions))

print("\nConfusion Matrix:")
print(confusion_matrix(y_val, nb_predictions))
```

```
Gaussian Naive Bayes Results
----------------------------
Accuracy: 0.4187568157033806
ROC-AUC: 0.652121762835075

Classification Report:
              precision    recall  f1-score   support

           0       0.20      0.92      0.33       142
           1       0.96      0.33      0.49       775

    accuracy                           0.42       917
   macro avg       0.58      0.62      0.41       917
weighted avg       0.84      0.42      0.46       917


Confusion Matrix:
[[131  11]
 [522 253]]
```

In [37]:

```python
from sklearn.model_selection import GridSearchCV, StratifiedKFold

# Create pipeline
nb_pipeline = Pipeline([
    ('preprocessor', preprocessor),

    ('to_dense', FunctionTransformer(
        lambda x: x.toarray() if hasattr(x, 'toarray') else x
    )),

    ('classifier', GaussianNB())
])

# Hyperparameter grid
param_grid = {
    'classifier__var_smoothing': np.logspace(-12, 0, 25)
}

# Stratified cross-validation
cv = StratifiedKFold(
    n_splits=5,
    shuffle=True,
    random_state=42
)

# Grid search
nb_grid = GridSearchCV(
    estimator=nb_pipeline,
    param_grid=param_grid,
    scoring='roc_auc',
    cv=cv,
    n_jobs=2,
    verbose=1
)

# Train
```

```
nb_grid.fit(X_train, y_train)

print("Best Parameters:")
print(nb_grid.best_params_)

print(
    "Best Cross-Validation ROC-AUC:",
    nb_grid.best_score_
)

# Best model
best_nb_model = nb_grid.best_estimator_

# Validation predictions
best_nb_probs = best_nb_model.predict_proba(X_val)[:, 1]

print(
    "Optimized Naive Bayes Validation ROC-AUC:",
    roc_auc_score(y_val, best_nb_probs)
)
```

```
Fitting 5 folds for each of 25 candidates, totalling 125 fits
Best Parameters:
{'classifier__var_smoothing': np.float64(1e-09)}
Best Cross-Validation ROC-AUC: 0.6597039982750623
Optimized Naive Bayes Validation ROC-AUC: 0.652121762835075
```

In [ ]:

```
from sklearn.feature_selection import SelectKBest, f_classif
from sklearn.model_selection import GridSearchCV

# Naive Bayes with feature selection
feature_selection_nb = Pipeline([
    ('preprocessor', preprocessor),

    ('to_dense', FunctionTransformer(
        lambda x: x.toarray() if hasattr(x, 'toarray') else x
    )),

    ('selection', SelectKBest(
        score_func=f_classif,
        k=100
    )),

    ('classifier', GaussianNB())
])

# Parameters
param_grid_fs = {
    'selection__k': [25, 50, 100, 200, 300, 'all'],

    'classifier__var_smoothing': [
        1e-2,
        1e-4,
        1e-6,
        1e-8,
        1e-10
```

```
    ]
}

# Grid search
nb_fs_grid = GridSearchCV(
    estimator=feature_selection_nb,
    param_grid=param_grid_fs,
    scoring='roc_auc',
    cv=5,
    n_jobs=2,
    verbose=1
)

# Train
nb_fs_grid.fit(X_train, y_train)

print("Best Parameters:")
print(nb_fs_grid.best_params_)

print(
    "Best Cross-Validation ROC-AUC:",
    nb_fs_grid.best_score_
)

# Best estimator
best_fs_nb = nb_fs_grid.best_estimator_

# Validation probabilities
fs_nb_probs = best_fs_nb.predict_proba(X_val)[:, 1]

print(
    "Feature Selection Naive Bayes ROC-AUC:",
    roc_auc_score(y_val, fs_nb_probs)
)
```

```
Fitting 5 folds for each of 30 candidates, totalling 150 fits
```

```
/opt/miniconda3/envs/section_5/lib/python3.14/site-packages/sklearn/feature_
selection/_univariate_selection.py:110: UserWarning: Features [  0  13  38
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selection/_univariate_selection.py:111: RuntimeWarning: invalid value encoun
tered in divide
  f = msb / msw
Best Parameters:
{'classifier__var_smoothing': 1e-08, 'selection__k': 25}
Best Cross-Validation ROC-AUC: 0.7359806827910427
Feature Selection Naive Bayes ROC-AUC: 0.7725942753293957
/opt/miniconda3/envs/section_5/lib/python3.14/site-packages/sklearn/feature_
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tered in divide
  f = msb / msw
```

In [ ]:
```python
#### Compare Gaussian Naive Bayes, Optimized Naive Bayes, and Feature Select

results_nb = pd.DataFrame({
    'Model': [
        'Gaussian Naive Bayes',
        'Optimized Gaussian NB',
        'Feature Selection NB'
    ],

    'Validation ROC-AUC': [
        roc_auc_score(y_val, nb_probs),
        roc_auc_score(y_val, best_nb_probs),
        roc_auc_score(y_val, fs_nb_probs)
    ]
})

results_nb = results_nb.sort_values(
    by='Validation ROC-AUC',
    ascending=False
)

print(results_nb)
```

```
                    Model  Validation ROC-AUC
2   Feature Selection NB            0.772594
0   Gaussian Naive Bayes            0.652122
1  Optimized Gaussian NB            0.652122
```

In [ ]:

```python
# Select the best Naive Bayes model using validation ROC-AUC

model_scores = {
    'GaussianNB': roc_auc_score(y_val, nb_probs),
    'Optimized GaussianNB': roc_auc_score(y_val, best_nb_probs),
    'Feature Selection GaussianNB': roc_auc_score(y_val, fs_nb_probs)
}

models = {
    'GaussianNB': nb_model,
    'Optimized GaussianNB': best_nb_model,
    'Feature Selection GaussianNB': best_fs_nb
}

best_model_name = max(model_scores, key=model_scores.get)

final_nb_model = models[best_model_name]

print("Selected Model:", best_model_name)
print("Validation ROC-AUC:", model_scores[best_model_name])


# Align test features with training columns
test_1 = test.reindex(columns=X_train.columns)


# Retrain selected pipeline using all available labeled training data
# X and y should represent the full training dataset
final_nb_model.fit(X, y)


# Predict test probabilities
test_probs = final_nb_model.predict_proba(test_1)[:, 1]


# Load Kaggle sample solution
sample_submission = pd.read_csv(
    "sample_solution.csv"
)


# Create submission
submission = sample_submission.copy()

submission["output"] = test_probs


# Save submission
submission.to_csv(
    "naive_bayes_submission.csv",
    index=False
```

```
)


print(submission.head())
print("Submission shape:", submission.shape)
```

```
Selected Model: Feature Selection GaussianNB
Validation ROC-AUC: 0.7725942753293957
   id        output
0   0  9.999877e-01
1   1  8.054762e-06
2   2  1.942114e-06
3   3  9.102362e-07
4   4  9.999904e-01
Submission shape: (1732, 2)
```

```
/opt/miniconda3/envs/section_5/lib/python3.14/site-packages/sklearn/feature_
selection/_univariate_selection.py:110: UserWarning: Features [  0  13  38
47  56  71  72  73  80  85  86  90  91  92  95  96  97  98
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tered in divide
  f = msb / msw
```

In [3]:
```
#### XGBoost Classifier with SHAP #####

##Here, we use SHAP (SHapley Additive exPlanations) to identify the most inf
```

In [ ]:
```
#Importing necessary libraries
#Code generated with the help of AI

import pandas as pd
import numpy as np

from sklearn.model_selection import cross_val_score
from sklearn.metrics import accuracy_score
from sklearn.metrics import f1_score
from sklearn.metrics import roc_curve, auc

from sklearn.model_selection import GridSearchCV

from sklearn.model_selection import train_test_split

import seaborn as sns
import matplotlib.pyplot as plt
from matplotlib.colors import ListedColormap
%matplotlib inline

from numpy import loadtxt
```

```
import xgboost as xgb

from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC


from sklearn.ensemble import (RandomForestClassifier, AdaBoostClassifier,
                              GradientBoostingClassifier, ExtraTreesClassifi
from sklearn.model_selection import KFold

from sklearn.naive_bayes import GaussianNB

import lightgbm as lgb
from sklearn.metrics import mean_squared_error

import matplotlib as mpl
from matplotlib import colors
from scipy import linalg
from sklearn import datasets
from sklearn.decomposition import PCA
from sklearn.discriminant_analysis import LinearDiscriminantAnalysis

from sklearn.model_selection import train_test_split, learning_curve
from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler, OneHotEncoder

from sklearn.dummy import DummyClassifier
from sklearn.linear_model import LogisticRegression

from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    roc_auc_score,
    classification_report,
    confusion_matrix
)
```

In [ ]:
```
#Train and test data

import pandas as pd

train = pd.read_csv('./train_data.csv')
test = pd.read_csv('./test_data.csv')

train.output.value_counts()
```

```
output
1    3873
0     711
Name: count, dtype: int64
```

In [ ]:
```
# Read and load the training and test data
train = pd.read_csv("train_data.csv")
test = pd.read_csv("test_data.csv")

try:
    sample_solution = pd.read_csv("sample_solution.csv")
except:
    sample_solution = None

print("Train shape:", train.shape)
print("Test shape :", test.shape)

display(train.head())
```

```
Train shape: (4584, 670)
Test shape : (1732, 669)
```

	x1	x2	x3	x4	x5	x6	x7	x8	x9	x10
0	0	0	0	1	-323.106605	2.694366	-1.987520	261.092784	0.013704	0.000100
1	0	0	0	1	-376.084691	0.969696	-6.933765	355.311648	0.030292	-0.000153
2	0	0	0	0	91.955425	2.621643	-2.581162	51.357206	0.036668	-0.000104
3	0	0	0	1	-391.814586	1.866914	-2.510799	382.900317	0.007947	-0.000028
4	0	0	0	0	-363.823732	2.951346	-3.726368	330.527539	0.010074	-0.000004

5 rows × 670 columns

In [ ]:
```
# Training Data information
train.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 4584 entries, 0 to 4583
Columns: 670 entries, x1 to output
dtypes: float64(474), int64(194), str(2)
memory usage: 23.4 MB
```

In [ ]:
```
# Identify numerical and categorical columns

print("\nColumns:")
print(train.columns.tolist())

# Missing values:
print("\nMissing values:")
display(train.isnull().sum().sort_values(ascending=False).head(20))

# Unique values
print("\nUnique values:")
display(train.nunique().sort_values())
```

Columns:
['x1', 'x2', 'x3', 'x4', 'x5', 'x6', 'x7', 'x8', 'x9', 'x10', 'x11', 'x12',
'x13', 'x14', 'x15', 'x16', 'x17', 'x18', 'x19', 'x20', 'x21', 'x22', 'x23',
'x24', 'x25', 'x26', 'x27', 'x28', 'x29', 'x30', 'x31', 'x32', 'x33', 'x34',
'x35', 'x36', 'x37', 'x38', 'x39', 'x40', 'x41', 'x42', 'x43', 'x44', 'x45',
'x46', 'x47', 'x48', 'x49', 'x50', 'x51', 'x52', 'x53', 'x54', 'x55', 'x56',
'x57', 'x58', 'x59', 'x60', 'x61', 'x62', 'x63', 'x64', 'x65', 'x66', 'x67',
'x68', 'x69', 'x70', 'x71', 'x72', 'x73', 'x74', 'x75', 'x76', 'x77', 'x78',
'x79', 'x80', 'x81', 'x82', 'x83', 'x84', 'x85', 'x86', 'x87', 'x88', 'x89',
'x90', 'x91', 'x92', 'x93', 'x94', 'x95', 'x96', 'x97', 'x98', 'x99', 'x10
0', 'x101', 'x102', 'x103', 'x104', 'x105', 'x106', 'x107', 'x108', 'x109',
'x110', 'x111', 'x112', 'x113', 'x114', 'x115', 'x116', 'x117', 'x118', 'x11
9', 'x120', 'x121', 'x122', 'x123', 'x124', 'x125', 'x126', 'x127', 'x128',
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8', 'x139', 'x140', 'x141', 'x142', 'x143', 'x144', 'x145', 'x146', 'x147',
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'x205', 'x206', 'x207', 'x208', 'x209', 'x210', 'x211', 'x212', 'x213', 'x21
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2', 'y3', 'y4', 'y5', 'y6', 'y7', 'y8', 'y9', 'y10', 'y11', 'y12', 'y13', 'y
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'y36', 'y37', 'y38', 'y39', 'y40', 'y41', 'y42', 'y43', 'y44', 'y45', 'y46',
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'z112', 'z113', 'z114', 'z115', 'z116', 'z117', 'z118', 'z119', 'z120', 'z12

```
1', 'z122', 'z123', 'z124', 'z125', 'z126', 'z127', 'z128', 'z129', 'z130',
'z131', 'z132', 'z133', 'z134', 'z135', 'z136', 'z137', 'z138', 'z139', 'z14
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7', 'z198', 'z199', 'z200', 'z201', 'z202', 'z203', 'z204', 'z205', 'z206',
'z207', 'z208', 'z209', 'z210', 'z211', 'z212', 'z213', 'z214', 'z215', 'z21
6', 'z217', 'z218', 'z219', 'z220', 'z221', 'z222', 'subject', 'phase', 'sta
te', 'output']

Missing values:
x1      0
y219    0
y221    0
y222    0
z1      0
z2      0
z3      0
z4      0
z5      0
z6      0
z7      0
z8      0
z9      0
z10     0
z11     0
z12     0
z13     0
z14     0
z15     0
y220    0
dtype: int64
Unique values:
x1         1
y57        1
y72        1
y73        1
y74        1
         ...
y60     4584
y61     4584
y62     4584
y19     4584
y161    4584
Length: 670, dtype: int64
```

In [ ]:
```python
##Checking the class distribution and imbalance in the training data

print("Class distribution:")
print(train['output'].value_counts())

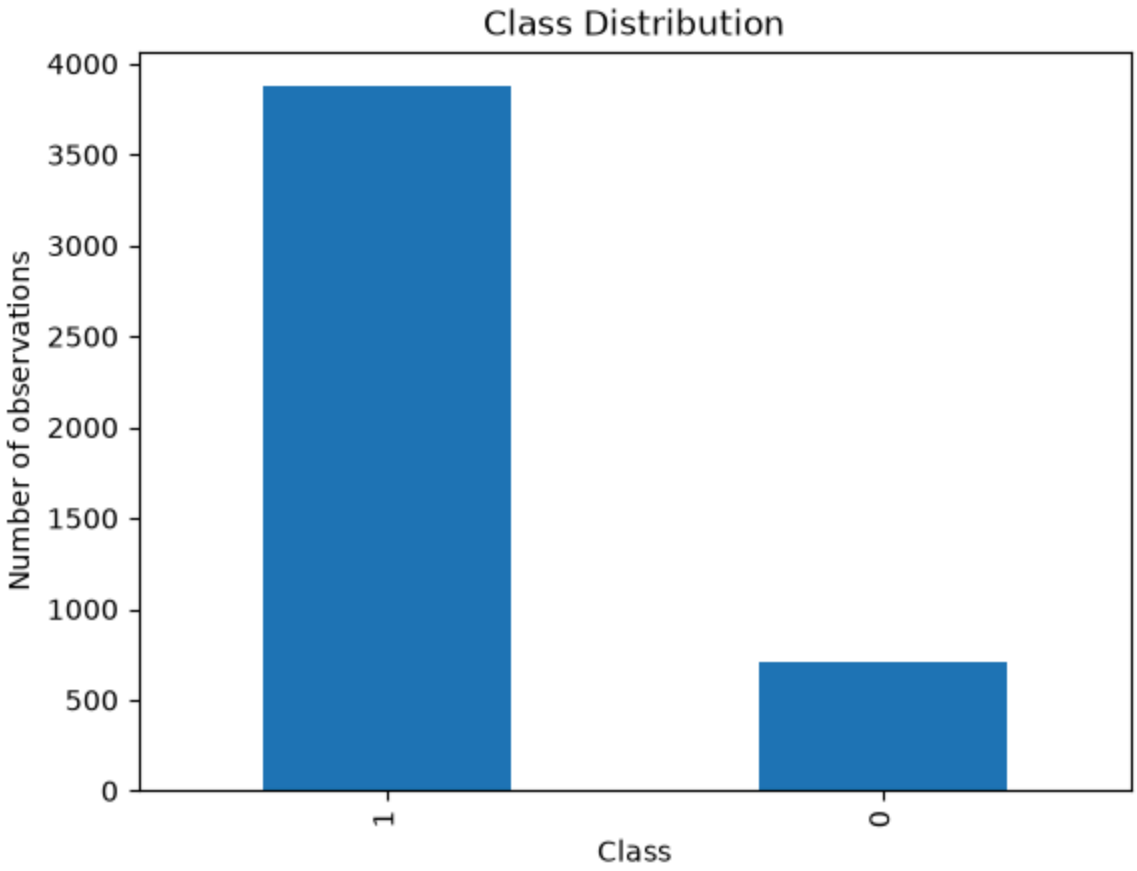
print("\nClass percentages:")
print(train['output'].value_counts(normalize=True) * 100)
```

```
train['output'].value_counts().plot(
    kind='bar',
    title='Class Distribution'
)

plt.xlabel("Class")
plt.ylabel("Number of observations")
plt.show()
```

```
Class distribution:
output
1    3873
0     711
Name: count, dtype: int64

Class percentages:
output
1    84.489529
0    15.510471
Name: proportion, dtype: float64
```



In [ ]:
```
# Separate features and target of the training data
X = train.drop(columns=['output'])
y = train['output']

# Stratified train-validation split
X_train, X_val, y_train, y_val = train_test_split(
    X,
    y,
```

```
    test_size=0.20,
    random_state=42,
    stratify=y
)

print("Training observations:", X_train.shape)
print("Validation observations:", X_val.shape)
```

```
Training observations: (3667, 669)
Validation observations: (917, 669)
```

In [ ]:
```
# Identify numerical and categorical columns
from sklearn.compose import ColumnTransformer

numeric_features = X.select_dtypes(
    include=['int64', 'float64']
).columns.tolist()

categorical_features = X.select_dtypes(
    include=['object', 'category', 'bool']
).columns.tolist()


# Numeric preprocessing
numeric_transformer = Pipeline([
    ('imputer', SimpleImputer(strategy='median')),
    ('scaler', StandardScaler())
])


# Categorical preprocessing
categorical_transformer = Pipeline([
    ('imputer', SimpleImputer(strategy='most_frequent')),
    ('encoder', OneHotEncoder(handle_unknown='ignore'))
])

# Combine preprocessing
preprocessor = ColumnTransformer([
    ('num', numeric_transformer, numeric_features),
    ('cat', categorical_transformer, categorical_features)
])
```

```
/var/folders/c3/1xdc_j3d5gnbhhnvrtwj5c540000gn/T/ipykernel_51303/3195068650.
py:8: Pandas4Warning: For backward compatibility, 'str' dtypes are included
by select_dtypes when 'object' dtype is specified. This behavior is deprecat
ed and will be removed in a future version. Explicitly pass 'str' to `includ
e` to select them, or to `exclude` to remove them and silence this warning.
See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#strin
g-migration-select-dtypes for details on how to write code that works with p
andas 2 and 3.
  categorical_features = X.select_dtypes(
```

In [ ]:
```
import numpy as np
import pandas as pd
import shap

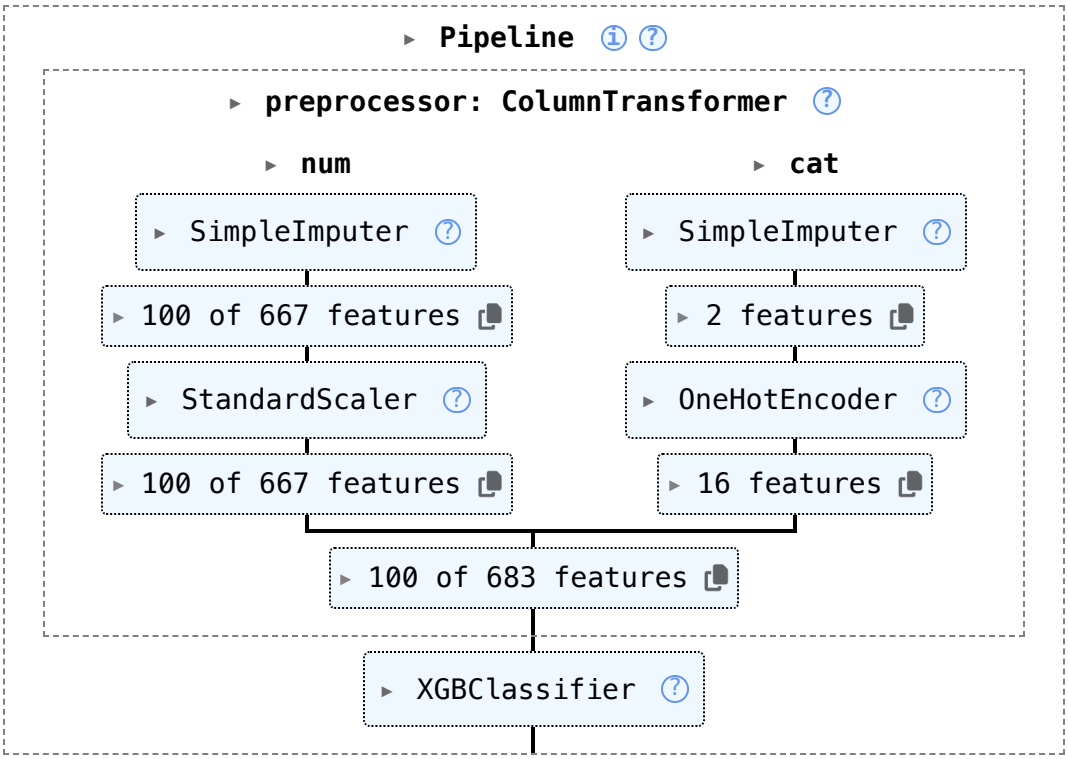
from xgboost import XGBClassifier
```

```
from sklearn.base import clone
from sklearn.pipeline import Pipeline
from sklearn.metrics import roc_auc_score
```

In [ ]:
```
#XGBoost Classifier

xgb_model = Pipeline([
    ('preprocessor', clone(preprocessor)),
    ('classifier', XGBClassifier(
        n_estimators=400,
        max_depth=3,
        learning_rate=0.03,
        subsample=0.8,
        colsample_bytree=0.8,
        reg_lambda=5,
        objective='binary:logistic',
        eval_metric='auc',
        tree_method='hist',
        random_state=42,
        n_jobs=2
    ))
])

xgb_model.fit(X_train, y_train)
```



In [ ]:
```
#Calculate the SHAP values for the XGBoost model

processor = xgb_model.named_steps['preprocessor']
classifier = xgb_model.named_steps['classifier']

X_train_processed = processor.transform(X_train)
X_val_processed = processor.transform(X_val)
```

```
explainer = shap.TreeExplainer(classifier)
shap_values = explainer.shap_values(X_train_processed)
shap_importance = np.abs(shap_values).mean(axis=0)
```

In [ ]:
```
#Select the top 50 features based on SHAP importance

top_features = np.argsort(shap_importance)[-50:]

X_train_shap = X_train_processed[:, top_features]
X_val_shap = X_val_processed[:, top_features]

print("Selected Features:", len(top_features))
```

```
Selected Features: 50
```

In [ ]:
```
#Train a new OptimizedXGBoost model using the selected features

shap_xgb = XGBClassifier(
    n_estimators=500,
    max_depth=3,
    learning_rate=0.025,
    subsample=0.85,
    colsample_bytree=0.85,
    min_child_weight=3,
    reg_alpha=1,
    reg_lambda=10,
    objective='binary:logistic',
    eval_metric='auc',
    tree_method='hist',
    random_state=42,
    n_jobs=2
)

shap_xgb.fit(X_train_shap, y_train)
```

▾ XGBClassifier ⓘ ⓘ

▸ Parameters

▸ Fitted attributes

In [ ]:
```
#Evaluate the ROC-AUC of the baseline XGBoost model and the SHAP-based XGBoo

baseline_preds = xgb_model.predict_proba(X_val)[:, 1]

shap_preds = shap_xgb.predict_proba(X_val_shap)[:, 1]

print("Baseline ROC-AUC:",
      roc_auc_score(y_val, baseline_preds))

print("SHAP XGBoost ROC-AUC:",
      roc_auc_score(y_val, shap_preds))
```

```
Baseline ROC-AUC: 0.8550749659245797
SHAP XGBoost ROC-AUC: 0.8469150386188097
```

In [ ]:
```python
#Code generated with the help of AI

# Fit preprocessing on all labeled data
final_processor = clone(preprocessor)

X_full = final_processor.fit_transform(X)

test_1 = test.reindex(columns=X.columns)
X_test = final_processor.transform(test_1)

# Recalculate SHAP feature importance using full training data
ranking_model = clone(classifier)
ranking_model.fit(X_full, y)

final_explainer = shap.TreeExplainer(ranking_model)
final_shap = final_explainer.shap_values(X_full)

final_importance = np.abs(final_shap).mean(axis=0)
final_top_features = np.argsort(final_importance)[-50:]

# Retrain final model
final_model = clone(shap_xgb)

final_model.fit(
    X_full[:, final_top_features],
    y
)

test_probs = final_model.predict_proba(
    X_test[:, final_top_features]
)[:, 1]
```

In [ ]:
```python
#Create the submission file for the XGBoost model with SHAP feature selectic

sample_solution = pd.read_csv(
    "sample_solution.csv"
)

submission = sample_solution.copy()

submission["output"] = test_probs

submission.to_csv(
    "xgboost_shap_submission.csv",
    index=False
)

print(submission.head())
print("Submission shape:", submission.shape)
```

```
   id    output
0   0  0.910014
1   1  0.917869
2   2  0.841883
3   3  0.781148
4   4  0.901500
Submission shape: (1732, 2)
```